

QVerse — Week 6: Entanglement and Bell States

Theory Booklet | Introduction to Quantum Computing & Programming

QVerse

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How this booklet fits the course

This theory booklet is designed to be read alongside `06_entanglement_and_bell_states.ipynb`. It contains the conceptual and mathematical material that should be explained during Week 6; the notebook then turns those ideas into experiments, circuit construction, and exercises.

Learning objectives

- Construct Bell states.
- Distinguish product states from entangled states.
- Analyze joint correlations from repeated measurement.
- Avoid interpreting entanglement as faster-than-light communication.

Recommended workflow

1. Read the relevant theory section.
2. Predict the result of the corresponding notebook experiment.
3. Run the notebook.
4. Explain any difference between the prediction and the output.
5. Complete the core exercises without treating the solution notebook as a substitute for reasoning.

1. What entanglement means

A multi-qubit pure state is entangled when it **cannot** be written as a product of separate states for the individual subsystems.

A product state has the form

$$|\psi\rangle_A \otimes |\phi\rangle_B.$$

An entangled state does not admit such a factorization.

The most important two-qubit example is the Bell state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

There is no choice of single-qubit states $|\psi\rangle$ and $|\phi\rangle$ whose tensor product equals $|\Phi^+\rangle$.

2. Creating a Bell pair

Start from

$$|00\rangle.$$

Apply H to the first qubit:

$$|00\rangle \rightarrow \frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

using the abstract ordering where the transformed qubit is written first.

Then apply a controlled- X with the first qubit as control and the second as target:

$$|00\rangle \rightarrow |00\rangle, \quad |10\rangle \rightarrow |11\rangle.$$

Therefore,

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

The key point is that the CNOT acts coherently on the superposition rather than choosing one branch classically.

3. Controlled- X gate

The controlled- X gate performs X on the target only when the control is 1.

Its action on the computational basis is

$$|00\rangle \rightarrow |00\rangle,$$

$$|01\rangle \rightarrow |01\rangle,$$

$$|10\rangle \rightarrow |11\rangle,$$

$$|11\rangle \rightarrow |10\rangle,$$

for the convention where the first written bit is the control.

It is a reversible two-qubit gate and is one of the standard entangling primitives.

4. The four Bell states

The Bell basis consists of

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}},$$

$$|\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}},$$

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}},$$

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}.$$

They are mutually orthogonal and maximally entangled.

The $+$ or $-$ sign changes a relative phase. As with one-qubit states, this phase may be invisible in one measurement basis and visible in another.

5. Z-basis correlations

For $|\Phi^+\rangle$,

$$P(00) = P(11) = \frac{1}{2},$$

while

$$P(01) = P(10) = 0.$$

Each individual qubit appears random if considered alone, but the pair is perfectly correlated.

For $|\Psi^+\rangle$, the outcomes are anti-correlated:

$$P(01) = P(10) = \frac{1}{2}.$$

These joint correlations are the first operational signature of the Bell states.

6. Why each subsystem can look random

The full Bell state is a pure state, but each individual qubit has no pure statevector description on its own.

Using density matrices, the full state is

$$\rho = |\Phi^+\rangle\langle\Phi^+|.$$

Tracing out one qubit gives the reduced state of the other:

$$\rho_A = \text{Tr}_B(\rho) = \frac{1}{2}I.$$

This is the maximally mixed one-qubit state.

So:

- the two-qubit system is in a perfectly known pure state;
- each subsystem individually is maximally uncertain.

This is a specifically quantum feature of entanglement.

7. Density-matrix idea

For a pure state $|\psi\rangle$,

$$\rho = |\psi\rangle\langle\psi|.$$

A density matrix allows both pure states and statistical mixtures to be represented.

For the maximally mixed qubit,

$$\rho_{\text{mix}} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}I.$$

The notebook uses a partial trace to show that the subsystem of a Bell pair has this form.

You do not need the full density-matrix formalism yet. The important interpretation is that subsystem states of an entangled system can be mixed even when the total state is pure.

8. X-basis correlations

The Bell states are easier to distinguish conceptually when measured in more than one basis.

Recall

$$|0\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}}, \quad |1\rangle = \frac{|+\rangle - |-\rangle}{\sqrt{2}}.$$

For example,

$$|\Phi^+\rangle = \frac{|++\rangle + |--\rangle}{\sqrt{2}}.$$

So $|\Phi^+\rangle$ is also correlated in the X basis.

The phase-changed state $|\Phi^-\rangle$ has a different X-basis correlation pattern. This is why the relative sign between Bell-state components is physically meaningful.

9. Correlation is not enough to prove entanglement

A purely classical correlated mixture could produce 00 half the time and 11 half the time:

$$\rho_{\text{classical}} = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11|.$$

This reproduces the same Z-basis counts as $|\Phi^+\rangle$.

To distinguish the coherent Bell state from that classical mixture, measurements in additional bases are required. More rigorous entanglement tests use entanglement witnesses or Bell inequalities.

For this course, the practical lesson is:

a single correlated histogram is suggestive but does not by itself demonstrate quantum entanglement.

10. Entanglement and no-signalling

Suppose Alice and Bob share a Bell pair. If Alice measures her qubit, her outcome is random. Bob's local outcome distribution is also random.

Alice cannot choose whether her result will be 0 or 1. Therefore she cannot use the correlation to encode a controllable faster-than-light message.

Only after Alice and Bob compare classical data do the correlations become visible.

This is the no-signalling principle in operational form.

11. Product superposition versus entangled state

Compare

$$|+\rangle|+\rangle = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}$$

with

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Both are superpositions. The first factors into independent one-qubit states; the second does not.

Measurements that probe correlations in multiple bases can separate these two cases.

12. Why entanglement is useful

Entanglement is a resource for:

- quantum teleportation;
- superdense coding;
- quantum error correction;
- quantum cryptographic protocols;
- distributed quantum information tasks;
- many quantum algorithms and simulation methods.

However, entanglement by itself is not an algorithm. It must be combined with a circuit structure and measurement strategy.

13. Common misconceptions

“Entanglement means one particle instantly sends a message to another”

No controllable message is transmitted by local measurement alone.

“Random individual outcomes mean the pair has no information”

The information can be contained in correlations.

“Any correlated data proves entanglement”

Classical correlations can mimic some measurement statistics.

“A Bell state is just two qubits that happen to match”

The key is the coherent nonseparable joint state, not only matching classical outcomes.

14. What to know before the notebook

You should be able to write all four Bell states and explain:

- why $H + CX$ creates $|\Phi^+\rangle$;
- why the pair can be pure while each subsystem looks mixed;
- why Bell states show characteristic correlations;
- why multiple measurement bases are useful;
- why entanglement does not allow faster-than-light communication.

Suggested references

- IBM Quantum Learning, *Basics of Quantum Information* and current Qiskit learning material: <https://quantum.cloud.ibm.com/learning/en/courses>
- IBM Quantum Documentation and Qiskit guides: <https://quantum.cloud.ibm.com/docs>
- Quantum Rings, *Quantum 101*: <https://www.quantumrings.com/course/quantum101>
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press.

The software-facing details in the practical notebook should be treated as the canonical implementation for this course. The purpose of this booklet is to explain the physics, mathematics, and algorithmic reasoning behind that implementation.